Smart Agricultural Technology 11 (2025) 100983

Contents lists available at ScienceDirect

Smart Agricultural Technology

journal homepage: www.journals.elsevier.com/smart-agricultural-technology

Feature-adaptive anomaly detection model for onion inspection system

Ziwei Song [a], Prawit Buayai [b], Koji Makino [b], Xiaoyang Mao [b,*]

[a] Department of Computer Science and Engineering, Faculty of Engineering, University of Yamanashi, Japan
[b] Graduate Faculty of Interdisciplinary Research, University of Yamanashi, 4-3-11 Takeda, Kofu, Yamanashi 400-8511 Japan

ARTICLE INFO

Keywords:
Smart agriculture
Onion detection
Anomaly detection
Inspection system
Deep neural network

ABSTRACT

Traditional industrial onion inspection relies heavily on manual labor, making it both time-consuming and prone to human error. We propose a novel automated onion inspection system comprising a **Vision-Based Inspection Module (VBIM)** for detecting defective onions and a **Defective Onion Filtering Module (DOFM)** for their physical removal. The VBIM introduces a new feature-adaptive anomaly detection method, which optimizes feature layer weights in a student-teacher anomaly detection model. This approach enhances the model's ability to focus on common features of normal onions while disregarding irrelevant variations, improving the discrimination of defective onions. The proposed system achieved **97.02 % accuracy** on a test dataset and **86.37 % accuracy** in real-world inspections. With an average processing speed of **4 FPS**, it demonstrates strong potential for real-time, high-precision defect detection in industrial applications.

1. Introduction

Onions are one of the most important vegetables globally. Their unique chemical composition, special taste, medicinal value, and long shelf life make them a crop of significant strategic importance and economic value. Not only are onions a vital component of the dietary cultures in many countries, but they are also widely utilized in the food industry and pharmaceutical sectors. Rich in nutrients such as vitamins, minerals, and antioxidants, onions play a crucial role in the global food supply. Furthermore, their excellent storability and transportability have made onions a key commodity in international agricultural trade. According to the Food and Agriculture Organization of the United Nations [1], the global import and export value of onions reached $8.3 billion, with a production volume of 110 million tons.

During industrial production, onions undergo multiple processes, including harvesting, cleaning, filtering, and transportation. The period from harvest to market sale often spans 1 to 2 months, during which the onions are predominantly stored in cold warehouses. Introducing defective or nearly rotten onions into storage can have disastrous consequences, as pathogens from rotted onions can rapidly spread within the facility, potentially destroying all healthy onions. The presence of defective onions accelerates disease transmission, leading to the swift proliferation of onion diseases and significant economic losses. Therefore, effective detection and removal of defective onions are crucial throughout the industrial production process.

In traditional industrial production methods, washed onions are manually inspected by experienced workers for damage and disease. However, this method has several limitations: variability in workers' subjective judgments leads to inconsistent results, fatigue and distraction increase error rates, and labor shortages, exacerbated by aging populations, drive up costs [2,3]. Misdiagnosis of diseased onions can result in significant losses in yield, time, money, and product quality. Moreover, onions are highly susceptible to diseases after harvest, such as sour skin and neck rot [4]. Disease infections often begin on the outer scales of onions, and early symptoms might be latent on the onion surface [5]. To address these issues, some commercial onion inspection systems have adopted hyperspectral imaging for non-destructive and high-precision defect detection [6,7]. Many of these systems also integrate robotic arms to physically remove defective onions from the production line. While such configurations offer high accuracy and full automation, they significantly increase the system's cost, complexity, and space requirements. As a result, these solutions are typically inaccessible to small and medium-sized enterprises. Consequently, there is an urgent need for a cost-effective and easy-to-deploy automated solution that can reliably detect and filter defective onions under real-world industrial conditions.

With the advancement of digital technology, particularly the development of artificial intelligence (AI), AI applications have become increasingly widespread in various fields [8-10]. It is common to utilize computer vision and deep learning techniques to address challenges in

* Corresponding author.
E-mail address: mao@yamanashi.ac.jp (X. Mao).

https://doi.org/10.1016/j.atech.2025.100983
Received 20 March 2025; Received in revised form 17 April 2025; Accepted 1 May 2025
Available online 1 May 2025
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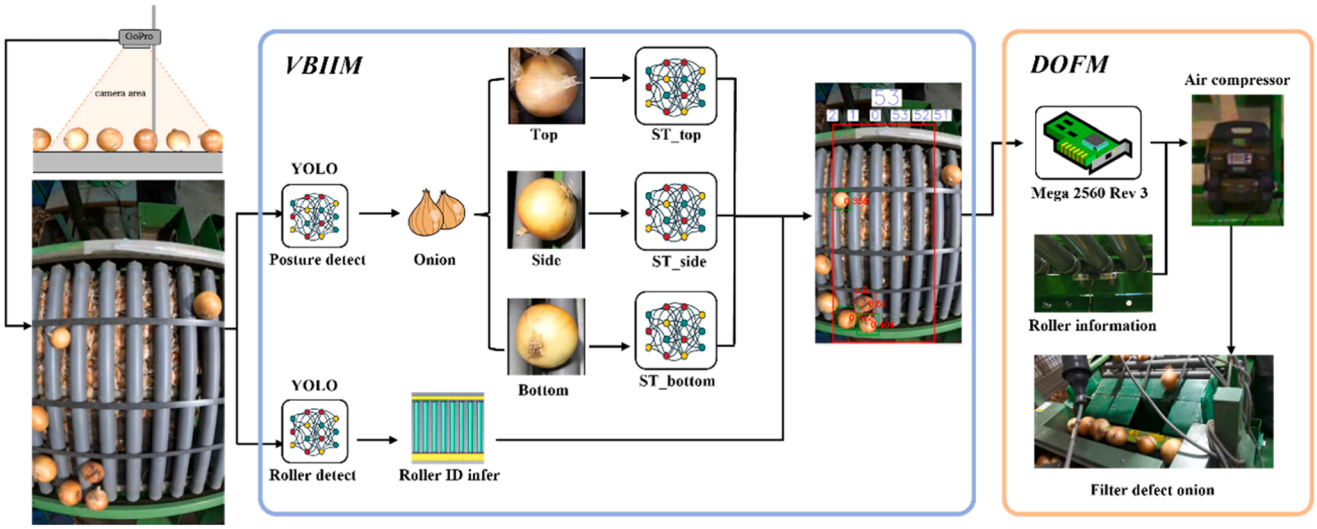
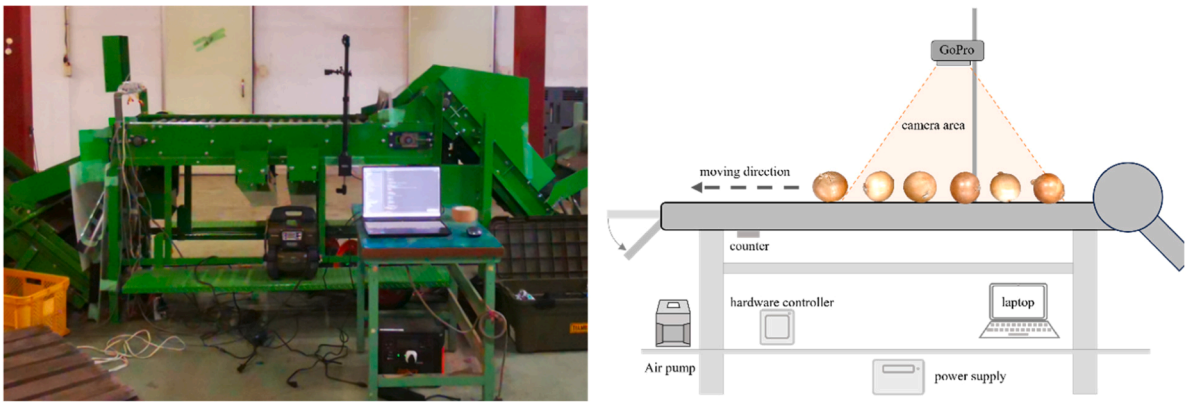


Fig. 1. Outline of the proposed approach.

agriculture [11-13]. Oppenheim et al. [14] proposed a potato disease classification algorithm based on an improved VGG network that could more accurately and quickly identify and classify spots on potato fruits. Dhingra et al. [15] proposed a neutral logic segmentation technique based on an extended fuzzy set, which achieves high-precision detection and classification of leaf diseases by extracting various leaf features. Qi et al. [16] improved the model's ability to identify and extract features related to tomato diseases by converting images from the RGB color space to the HSV color space and optimizing the model structure. Du et al. [17] detect spikelets for precision mechanical thinning in table grape inflorescence under complex environments with a deep learning model. In the context of onion classification, several studies have explored AI-based methods for defect detection and classification. As early as 2002, neural network technology combined with traditional image processing techniques were used to detect and classify defects in sweet onions [18]. Rodriguez Galdon et al. [19] classified the characteristics of onions of different colors using cluster analysis and neural networks. Wang and Li [20] introduced an RGB-d-based approach for measuring the size attributes of onions through point cloud imaging. Wang and Li [21] designed and developed a multimodal machine vision system that integrates multiple imaging technologies such as spectroscopy, X-ray, and 3D depth imaging, enabling non-destructive evaluation of onion quality through comprehensive image analysis. Building upon convolutional neural networks (CNNs), Puspadhani et al. [22] developed an onion recognition machine based on the visual features of different onion varieties. Similarly, Waghmare et al. [23] employed CNNs and HSV imaging to classify red and white onions. Despite the progress, most of these approaches rely on classification-based models trained on RGB images, which face notable limitations in industrial applications. These models typically require large and well-balanced datasets, with sufficient labeled samples representing both normal and various types of defective onions. However, such data conditions are rarely met in real-world industrial settings. In practice, normal onions are consistently abundant, whereas defective onions occur infrequently and display highly diverse and unpredictable appearances. This natural imbalance between classes poses a significant challenge for constructing comprehensive and balanced training datasets. Consequently, classification models trained under such conditions often struggle to generalize to rare or previously unseen defects in practice.

To address these limitations, we adopt an anomaly detection approach. This type of model is trained exclusively on normal onion images, which are abundant and easy to collect in industrial production lines. By learning the intrinsic feature distribution of normal samples, the model can effectively detect abnormal onions as deviations from this learned pattern. This greatly reduces the need for exhaustive defect labeling and enhances the model's ability to generalize to new, unseen defect types. Recent studies have demonstrated that anomaly detection methods can achieve over 90 % accuracy in industrial scenarios [24].



(a) Onion inspection system. (b) Components of system.

Fig. 2. The structure of the onion inspection system.

However, determining an appropriate threshold for identifying anomalies remains challenging in practice, due to the inherent variability within normal data under real-world industrial conditions.

In this paper, we propose a lightweight and affordable feature-adaptive anomaly detection method that optimizes weight parameters across different feature layers of the student teacher model to minimize the variance of normal samples. This enhancement improves the robustness and adaptability of anomaly detection algorithm in diverse industrial environments. By integrating this enhanced anomaly detection method with the You Only Look Once (YOLO) detection architecture [25], we developed an onion inspection system that is simple, effective, and easy to apply in real-world settings. Specifically, the YOLOv8 [26] model was selected for its ability to significantly enhance detection speed while maintaining high detection accuracy, making it one of the most effective algorithms for object detection. Our proposed method optimizes the detection framework to maintain real-time processing speeds while improving recognition accuracy and efficiency. This integrated system leverages the complementary strengths of anomaly detection and YOLO-based object detection to enable automated detect defective onions and filter them out in industrial production. By combining high-speed processing with reliable accuracy, the proposed system ensures efficient, low-cost, and robust detection in real-world scenarios. This research lays a solid foundation for future advancements in intelligent agricultural automation.

2. Materials and methods

Fig. 1 illustrates the workflow of the proposed automated onion inspection system, which integrates anomaly detection and defective onion filtering. This system leverages multiple models to enable comprehensive onion inspection, combining detection, classification, and filtering into a unified process. Detailed descriptions of the system's components and design are presented in the subsequent sections.

2.1. Inspection system design

The proposed onion inspection system consists of two main components: Vison-Based Anomaly Inspection Module (VBIIM) and Defective Onion Filtering Module (DOFM), as illustrated in Fig. 2. This integrated design enables precise onion detection and filtering, offering a scalable and efficient solution for industrial applications.

The VBIM plays a central role in ensuring accurate onion detection and quality assessment. This module uses a multi-model collaborative approach to process images captured by a GoPro camera as onions move along the conveyor rollers. These images, containing structural details and color information, are analyzed using a combination of YOLO object detection and anomaly detection algorithms. The YOLO model is responsible for locating each onion in the image by generating bounding boxes, which are then used to crop individual onion regions. These cropped images are subsequently passed to the anomaly detection module, which evaluates their visual consistency with known normal samples and assigns an anomaly score used for defect classification.

By employing transfer learning on a pre-trained YOLO model, VBIM segments onion images and determines their positions on the conveyor rollers. Simultaneously, segmented images are processed by anomaly detection models optimized using the Multi-Resolution Weight method. This enhances the accuracy and reliability of the analysis, enabling real-time image processing and precise identification of defective onions by comparing their anomaly scores against a pre-set threshold.

The DOFM manages the mechanical operations of defective onion filters. It includes components such as flappers and air compressors for removing defective onions, along with a counter unit to track onion positions. Seamless communication between VBIM and DOFM via serial communication, enabling efficient data and command transmission. Guided by the ID information from the VBIM, the DOFM automatically activates the air pump and flappers to sort and filter out abnormal onions with precision.

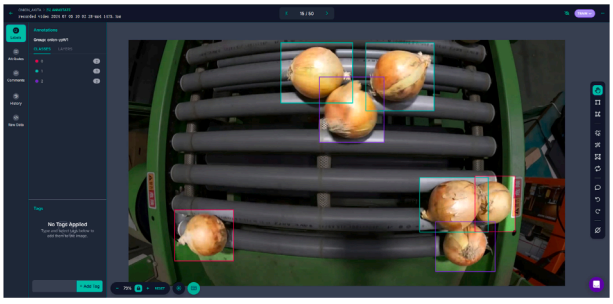


Fig. 3. Label fabrication.

The VBIM was developed in Python, leveraging deep learning libraries for image processing and model inference, while the DOFM is programmed in *C++* for efficient operation.

During operation, onions move along the conveyor rollers, passing sequentially through the camera capture zone and the flappers. A specific distance is maintained between the camera and the flappers, providing sufficient time for the system to process images and perform AI-based anomaly detection. Specifically, the YOLO model detects and identifies each onion on the conveyor rollers, determining their positions and segmenting their images for quality evaluation by the anomaly detection model. If a defective onion is identified, its information is transmitted to the DOFM, which filters the defective onion when it reaches the end of transport belt.

By leveraging the precision of the YOLO model and the computational efficiency of GPU acceleration, the proposed system enables real-time, high-accuracy detection and efficient filtering of defective onions. This work provides a robust and cost-effective solution for defective onion inspection in industrial applications.

2.2. Image processing for onion inspection

To develop a fully automated onion inspection system capable of real-time and high-accuracy detection, we employed a multi-model approach to ensure precision across all stages of detection and classification. The system integrates five specialized models: two YOLO models and three anomaly detection models, each designed to handle distinct tasks such as onion detection, roller positioning, and onion quality classification.

The two YOLO models serve distinct purposes, one, onion detection model, is for detecting and locating onions within the images, extracting information for anomaly detection processing. Another, roller detection model, identifies the positions of the rollers, ensuring precise coordination with the onion detection results to maintain accurate onion placement on the conveyor system. For the models training, the YOLOv8s pre-trained model was adopted as a foundation and further optimized using transfer learning, which leverages knowledge from large-scale datasets, significantly reducing training time while improving performance on specific tasks. The anomaly detection models classify onions into two categories: "normal" and "abnormal", by comparing the anomaly scores to a predetermined threshold.

To train the onion and roller detection models, two large datasets were created: one for onion images and another for roller images. Each image in these datasets was carefully annotated to indicate the categories and precise locations of onions and rollers. The annotation process was conducted according to strict predefined standards to ensure accuracy and consistency. The annotation procedure is shown in Fig. 3.

For anomaly detection, the trained YOLO onion detection model automatically cropped onion regions from the images. The cropped images were categorized into normal onions, which represent high-quality onions used for model training, and abnormal onions, which refer to defective onions with issues such as scars, mold, abnormal

Fig. 4. Bad onion case.

shapes, or rot, used for testing. Examples of defective onions are shown in Fig. 4.

To ensure consistency and robustness in model performance, several preprocessing techniques were applied. Image padding was used to standardize the dimensions of the cropped onion images, resolving potential size inconsistencies and ensuring compatibility with the model input. Additionally, augmentation was employed to enhance dataset diversity and improve generalization by applying transformations such as flipping and mirroring. These techniques reduced overfitting risks and improved the model's performance in real-world scenarios.

To address potential size inconsistencies in the cropped images, we applied padding to standardize their dimensions. This preprocessing step not only prevented size mismatches during model input but also improved the model's robustness to variations in image size. Additionally, we applied data augmentation techniques to enhance the diversity and generalization capability of the dataset. Transformations such as flipping and mirroring were used to expand the onion detection dataset, reduce the risk of overfitting, and improve the model's performance in real-world scenarios.

Through this series of preprocessing steps, including image cropping, padding, and augmentation, we laid a solid foundation for model training and inference. These steps ensured the accuracy and efficiency of the system during practical operation.

2.3. System configuration and evaluation indicators

2.3.1. The configuration of vision-based inspection module

The VBIM is built upon a collaborative framework that combines advanced object detection and classification models. Specifically, it employs YOLO models for precise localization of onions and conveyor rollers, while anomaly detection models focus on assessing onion quality. These components work in tandem to facilitate seamless real-time detection and positioning operations, ensuring high efficiency and accuracy.

All models were trained using the PyTorch framework on a high-performance server equipped with an Intel processor (3.0 GHz base frequency), 125 GB RAM, two NVIDIA RTX A6000 GPUs, and CUDA 12.3. Once trained, the models were deployed to a laptop with an AMD processor (3.2 GHz base frequency), 40 GB RAM, and an NVIDIA RTX 3060 Laptop GPU.

2.3.2. The configuration of defective onion filtering module

The DOFM is designed to execute precise mechanical operations. It is centered on an Arduino Mega 2560 Rev 3 development board, which controls various components: Flapper mechanism, Roller counter, and Control box.

The Flapper mechanism consists of components such as flappers and air valve controllers, which are connected to an air pump via ducts. The air pressure is adjusted through the controller, enabling the Flapper to flip within a range of 0 to 90 degrees. An HAIGE air compressor is used to supply the required air pressure. The roller counter tracks the positions of rollers to ensure that the system provides accurate onion filtration. The control box protects the Arduino board and associated components, ensuring stability and safety during operation. Through precise programming, the Arduino board coordinates the flapper to remove defective onions based on detection results. The components of the DOFM are illustrated in Fig. 5.

2.3.3. Communication protocol between VBIM and DOFM

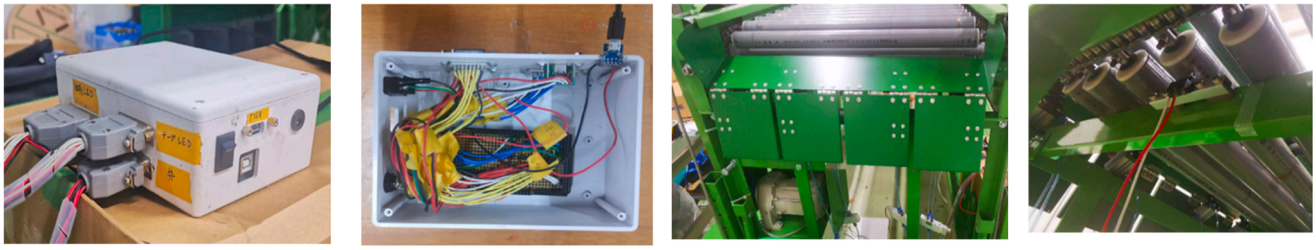
In the proposed system, the VBIM and the DOFM communicate via a serial protocol using a serial port, implemented through the QtSerial framework. Each message sent by the VBIM follows a structured format, beginning with a fixed start byte 255, followed by two comma-separated integers: the line ID and the onion ID. For example, a message such as 255,1,48 indicates that onion number 48 on line 1 has been identified as defective.

Upon receiving this message, the DOFM decodes the data and temporarily stores the onion ID into a defect cache buffer. As onions progress along the conveyor belt and pass through the filtering point, the roller system continuously updates the currently active onion ID. When a new roller ID is detected, the DOFM checks whether this ID exists in the defect cache. If the ID is found in the buffer, the system immediately triggers the actuator to drop the filtering barrier, thereby removing the defective onion from the processing line. If the ID is not found in the buffer, the onion is allowed to pass normally.

This communication and control mechanism ensures that filtering decisions are executed in real-time with minimal latency, while maintaining accuracy and synchronization between vision detection and mechanical response.

2.3.4. Evaluation indicators

In this experiment, the hyperparameters of the YOLOv8 model were uniformly set to ensure robust training and evaluation. The initial learning rate was set to 0.01, with a momentum value of 0.9. A batch



(a) Control box (outside). (b) Control box (inside). (c) Flapper. (d) Counter.

Fig. 5. The components of DOFM.

size for each training session was set to 8. The number of iterations was set to 400. For performance evaluation, the mean average precision (mAP) was adopted. The mAP metric is mathematically defined as:

$$mAP = \frac{1}{N}\sum_{n=1}^{N}\int_{0}^{1} P(R), dR \tag{1}$$

Where N is the number of categories of the test samples. $P(R)$ denotes precision as a function of recall, and the integral computes the area under the precision-recall curve.

For anomaly detection, a student-teacher model [27] was employed. The teacher model, trained on a large and diverse dataset, acts as a general feature extractor. The student model, with a simpler architecture, is trained using only normal data. It learns the distribution of normal instances by mimicking the teacher model's feature extraction capabilities. During validation, anomalies are determined by comparing the differing performances of the student and teacher models on anomalous data.

The student model, having never been exposed to anomalies and limited in capacity, struggles to extract features for anomalous inputs, resulting in noticeable discrepancies compared to the teacher model. These differences are quantified using a loss function, with a predefined threshold applied to identify anomalous samples. The area under the receiver operating characteristic curve (AUROC) was used as the primary metric for evaluating anomaly detection performance.

The Receiver Operating Characteristic (ROC) curve is generated by plotting the true positive rate (TPR) against the false positive rate (FPR) for a range of threshold values.

$$TPR = \frac{TP}{TP + FN} \tag{2}$$

$$FPR = \frac{FP}{FP + TN} \tag{3}$$

where TP is the number of true positives and FN is the number of false negative. FP represents the number of false positives, and TN is the number of true negative.

To generate the ROC curve, predicted anomaly scores are sorted in descending order. For each unique score, a threshold is applied to determine the classification as anomalous or normal. The corresponding TPR and FPR values are computed for each threshold, forming a sequence of points in the ROC space. The AUROC is then calculated as the integral of the ROC curve, which can be approximated using the trapezoidal rule for discrete data:

$$AUROC = \sum_{i=1}^{n-1}(FPR_{i+1} - FPR_i) \times \frac{TPR_{i+1} + TPR_i}{2} \tag{4}$$

Where n denotes the total number of thresholds, and $(FPR_i,\ \mathrm{TPR_i})$ represents the *FPR* and *TPR* at the *i th* threshold.

To provide a comprehensive evaluation of the system, we used accuracy as the overarching metric. It is defined as the percentage of correctly predicted instances out of the total number of images in the test set, calculated using the formula:

$$\text{Accuracy} = \frac{\text{Number of correctly predicted}}{\text{Number of images in test set}} \times 100 \tag{5}$$

2.4. YOLO model selection

The YOLO object detection algorithm, originally proposed by Redmon, introduced a regression-based approach that significantly improved detection speed while maintaining high accuracy compared to the R-CNN series of object detection networks. This advancement provides essential technical support for real-time detection tasks. YOLOv8, an enhanced version of YOLOv7 [28], incorporates advanced architectural features CSPNet [29] and an enhanced Anchor-Free structure, enabling faster and more accurate inference and detection.

YOLOv8 incorporates state-of-the-art backbone and neck architectures, enhancing feature extraction and object detection capabilities. It combines high accuracy with fast detection speed, making it one of the most reliable object detection algorithms available. The YOLOv8 GitHub repository offers five network variants: YOLOv8n (nano), YOLOv8s (small), YOLOv8m (medium), YOLOv8l (large), and YOLOv8x (extra-large). Among these, YOLOv8n is specifically designed for nano hardware, while the other four models are general-purpose. These variants differ in the depth and width of their architectures, with larger models offering higher accuracy at the cost of increased computational complexity.

Table 1
Comparison of YOLO model prediction results.

Models	mAPval 50–95	A100 TensorRT(ms)	Params (M)	FLOPs (B)
YOLOv8s	44.9	1.20	11.2	28.6
YOLOv8m	50.2	1.83	25.9	78.9
YOLOv8l	52.9	2.39	43.7	165.2
YOLOv8x	53.9	3.53	68.2	257.8

As shown in Table 1, the performance metrics of YOLOv8 models improve progressively from YOLOv8s to YOLOv8x in terms of feature extraction capability and detection accuracy. However, these improvements come at the cost of increased computational complexity and inference time. The YOLOv8s was selected as it achieved the best balance between speed and accuracy, meeting the real-time requirements of our system. While larger variants (e.g., YOLOv8x) offer higher accuracy, their significantly increased computational cost makes them less suitable for deployment on resource-constrained hardware.

In the deployed system, the YOLOv8s model plays a crucial role in detecting onions from continuous video input. Specifically, video frames are captured in real time using a GoPro camera mounted above the conveyor belt. The frame is processed by the YOLOv8s model to locate onions via bounding boxes. These bounding boxes are then used to crop the corresponding regions of the image, generating isolated onion images for further analysis. The cropped onion images are subsequently passed to the anomaly detection module. If the calculated anomaly score exceeds a predefined threshold, the onion is classified as defective and marked for removal by the downstream filtering mechanism.

Table 2
The performance of anomaly detection models on collected Saga onion dataset.

	Ganomaly [32]	ST	RD [33]	SimpleNet [24]
Onion data	93.2 %	**99.61 %**	97.2 %	99.46 %

2.5. Anomaly detection with feature-adaptive weight optimization

Anomaly detection is a widely used technique in industrial defect detection, which can detect defects and irregularities that are often unpredictable. In the context of onion harvesting, onions can suffer from damage and decay, with the locations and extent of defects being highly variable and difficult to predict. This makes anomaly detection an ideal approach for identifying defective onions, as it is designed to detect deviations from normal patterns. Common methods for anomaly detection include reconstruction-based approaches, embedding-based techniques, and knowledge distillation methods.

The performance of anomaly detection models on the Saga onion dataset (See Section 3.1 for the details on the dataset) is presented in Table 2. Among the models, the ST model [27] achieved the highest accuracy of 99.61 %. Due to its superior performance, we selected the ST model for anomaly detection in this study. It comprises two key components: a teacher model and a student model. The teacher model is a

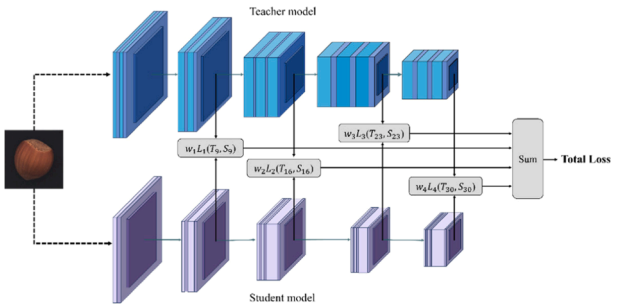


Fig. 6. The structure of weight optimized anomaly detection model.

VGG16 [30] network pre-trained on the ImageNet dataset [31], while the student model shares a similar structure with the teacher model but designed with lower complexity.

During training, the student model is guided by a loss function to mimic the teacher model's feature extraction capabilities. Importantly, the training process uses only normal data, enabling the student model to learn the patterns and characteristics of normal samples. As a result, the student model becomes highly specialized in recognizing normal patterns but lacks the ability to generalize to anomalous data.

During validation, when the student model is exposed to anomalous data, it struggles to extract relevant features due to its limited capacity and specialized training. Consequently, it fails to accurately represent irregular patterns. In contrast, the teacher model, pre-trained on a large and diverse dataset, retains a broad and robust feature extraction ability, enabling it to generate meaningful representations for both normal and anomalous samples.

The difference between the outputs of the teacher and student models is used to identify anomalies. If the discrepancy is below a predefined threshold, the sample is classified as normal; otherwise, it is considered anomalous. This approach effectively leverages the complementary strengths of the teacher and student models to distinguish normal onions from defective ones.

Many anomaly detection methods, including the ST model, focus on distinguishing between normal and abnormal features. The ST model achieves this by analyzing differences between the corresponding feature layers of the teacher and student models. However, these methods often overlook the natural variability inherent within normal data, which can obscure the boundaries between normal and abnormal features, making it challenging to differentiate them using simple threshold-based approaches.

To address this limitation, we propose a novel feature-adaptive weight optimization method. As shown in Fig. 6, the core idea is to optimize the weight distribution across different feature scales in the loss function. By assigning lower weights to feature scales with higher local variance in normal data, the method minimizes the influence of internal variability. This enables the model to concentrate on stable and consistent patterns, thereby enhancing its ability to differentiate between normal and abnormal features effectively.

Table 3
Workflow of the proposed feature-adaptive weight optimization for the ST model.

Input: normal training samples $train_{data}$, pre-trained teacher model $model_t$, student model $model_s$
Initialization: $w_{vi} = 1$, $w_{di} = 1$ and student model $model_s$
Setting parameters: convergence ε for stop loop
Loop (until $\Delta w_{vi} \leq \varepsilon$ and $\Delta w_{di} \leq \varepsilon$**)**
Train $model_s$ with $Loss = \sum_i [(w_{vi}L_{val}) + (w_{di}L_{dir})]$
Freeze $model_s$:
$w_{vi}', w_{di}' = \min_{w_{vi}, w_{di}} anoamly\ score(train_{data})$
$\Delta w_{vi} = |w_{vi}' - w_{vi}|$ and $\Delta w_{di} = |w_{di}' - w_{di}|$
Update w_{vi}' to w_{vi}, w_{di}' to w_{di}
Output: $model_s$, w_{vi} and w_{di}

The loss function in the ST model consists of two components: L_{val} and L_{dir}. The L_{val} measures the Euclidean distance between the corresponding layers of the teacher and student models. The L_{dir} evaluates cosine similarity between the corresponding layers to account for directional consistency. In our proposed method, these weights are dynamically optimized during training to ensure that the model focuses more on the similarities within the training dataset and less on the variations. The total losses function with weight w_{vi} and w_{di} is defined as

$$Loss = L_{val} + L_{dir} \tag{6}$$

$$L_{val} = \sum_{i}^{n} w_{vi} * (a_t(i) - a_s(i))^2 \tag{7}$$

$$L_{dir} = \sum_{i}^{n} w_{di} * \left(1 - \frac{a_t(i) \cdot a_s(i)}{\| a_t(i) \| \| a_s(i) \|}\right) \tag{8}$$

Where $a_t(i)$ is the activate results of the i *th* intermediate layer of teacher model and the $a_s(i)$ is the activate results of the i *th* intermediate layer of student model. $w_{vi}(i)$ is the Euclidean distance weight of the i *th* layer, and similarly, $w_{di}(i)$ is the weight of the cos similarity of the i *th* layer.

The workflow for obtaining the best weights in the ST model is detailed in Table 3. All weights were set to 1.0 during the first phase of training. After completing the initial training of the student model, the entire normal dataset is fed into the trained model to compute anomaly scores. These scores are aggregated to calculate the overall anomaly score for the normal dataset. Using this overall score as an optimization target, the weights are adjusted to minimize it. The optimization alternates between freezing the student model and updating the weights, and freezing the weights while retraining the student model. This alternating process continues until the weights stabilize.

By focusing on minimizing the overall anomaly score, the optimized weights prioritize feature scales with lower variance in normal data. This refinement enhances the model's ability to concentrate on stable and meaningful patterns, significantly improving its performance in anomaly detection tasks.

2.6. Precise roller ID mapping algorithm

Following the quality inspection phase, the next challenge in the onion inspection system is to ensure that defective onions are accurately filtered based on their inspection results. However, this process depends on correctly linking each onion to its corresponding position on the conveyor belt, a task complicated by the dynamic nature of the conveyor system. In this context, precise roller identification becomes critical, as the rollers not only transport the onions but also serve as unique positional markers that guide the filtering of defective onions.

To address this challenge, roller detection model and precise roller ID mapping algorithm were specifically designed to adapt to the unique structural and operational characteristics of the onion conveyor system. The roller detection model is used to detect and locate rollers within the camera's detection area, while the roller ID mapping algorithm ensures the precise assignment of roller IDs to accurately associate onions with their corresponding rollers.

In the onion inspection system, each onion is positioned between two rollers, and the roller on the moving side is designated as the onion's unique identifier (ID). To accurately identify defective onions and their corresponding roller IDs, the system calculates offsets during the detection process. First, a *camera_offset* is determined based on the distance between the camera and the counter. Similarly, a *counter_offset* is calculated based on the distance between the counter and the flapper.

During operation, the counter continuously transmits the IDs of rollers (*R_ID*) passing through its detection zone to the detection system

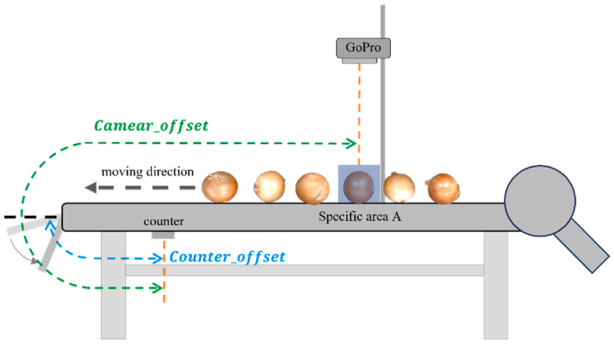


Fig. 7. Offset relationships in the inspection system.

via serial communication. As illustrated in Fig. 7, the roller ID corresponding to a specific area A within the camera's detection zone is calculated as:

$$A_ID = R_ID - camera_offset \tag{9}$$

Once A_ID is identified, the IDs of other rollers in the detection area can be inferred based on the results from the roller detection model. If an abnormal onion is detected, its associated roller ID is recorded as *Bad_onion_ID*. And the abnormal onion's ID is then adjusted using *counter_offset* to determine the final *final_ID* to be transmitted to the hardware system for removal:

$$final_ID = Bad_onion_ID + counter_offset \tag{10}$$

As shown in Fig. 8. the specific area A located directly below the camera, which is chosen for its minimal image distortion. Within this area, two scenarios can occur: one roller in area A, or two rollers in area A. When only one roller is in this area, the roller ID directly corresponds to that roller. However, when two rollers are present, a more nuanced approach is required to determine their IDs accurately.

To handle situations with two rollers, a precise roller ID mapping algorithm precise roller ID mapping algorithm is proposed, as outlined in Table 4. This algorithm assigns roller IDs by comparing their positions and IDs with the detected positions in the previous frame. By iteratively analyzing these data points, the algorithm ensures consistent and precise roller identification, enhancing the accuracy of onion filter.

Table 4
Proposed precise roller ID mapping algorithm.

Input: Real-time detection results of rollers, the ID of roller ID_i received from DOFM at each time step i.
Initialize the current time step i,
Setting parameters: Total number of frames to be processed N.
For $i = 1$ to N **do**
check the number of rollers n in specific area A .
If $n = 1$ **do**
Assign the ID_i to the roller.
If $n = 2$ **do**
Compare the ID_i with the previously recorded ID P_{id}
If $ID_i = P_{id}$ **do**
Assign ID_i to the roller that is farther from $P_{position}$
If $ID_i > P_{id}$ **do**
Assign ID_i to the roller that is closer to $P_{position}$
Record the roller position $P_{position}$ and ID P_{id}
Infer ID to other rollers.

3. Results and discussion

3.1. Data collection

In our experiments, we employed a GoPro Hero 10 Black camera to capture video data from onion processing lines at two different locations: Saga and Yurihonjō, Japan. The camera, with a resolution of 1920×1080, was positioned 30 cm above the conveyor roller to continuously record the onions. To ensure variability, we captured onion images from multiple angles. Each onion instance was annotated with its minimum bounding rectangle to minimize background noise and enhance detection accuracy. The camera setup is illustrated in Fig. 9.

As a preliminary step, we conducted a small-scale data collection in Saga, Saga Prefecture, Japan, to validate the accuracy of our proposed method. Video data were recorded at an onion processing line, and after manual filtering, 5000 images of normal onions and 350 images of abnormal onions were annotated. This dataset provided an essential baseline for evaluating the performance of our anomaly detection model before scaling up to larger datasets.

Following the successful validation of our method, a larger dataset was collected at a facility in Yurihonjō, Akita prefecture, Japan. The setup included a conveyor system with a camera positioned similarly to that in Saga. This collection aimed to create a more diverse and comprehensive dataset for robust model training. To train the YOLO onion detection model, we compiled a dataset consisting of 460 images containing 4681 annotated onion instances. Similarly, for the YOLO roller detection model, 500 roller images were used, yielding 2124 annotated roller instances. After training these models, the YOLO onion detection model was applied to the recorded video data to extract additional onion images. Following careful manual screening, we obtained a substantial dataset comprising 42,554 normal onion instances and 798 abnormal onion instances. These datasets were subsequently used for training and evaluating the anomaly detection model.

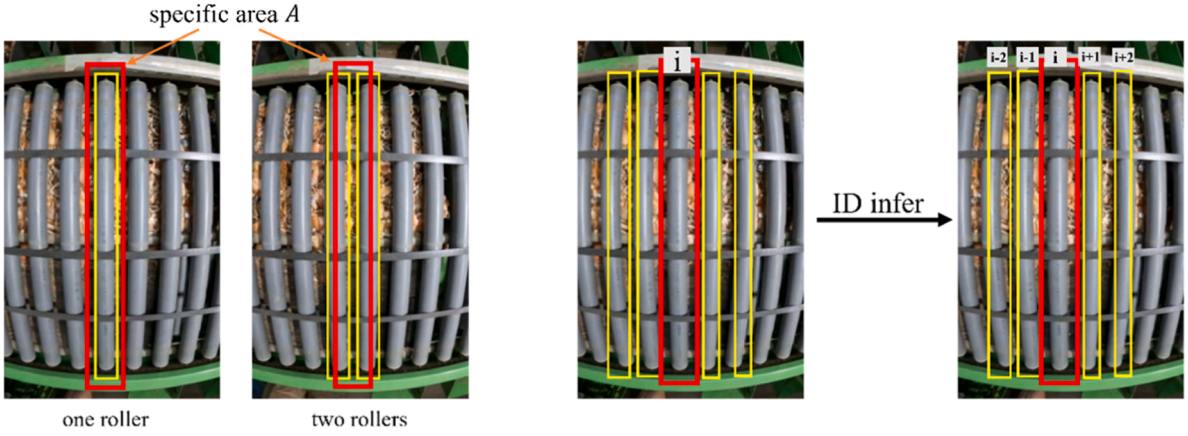
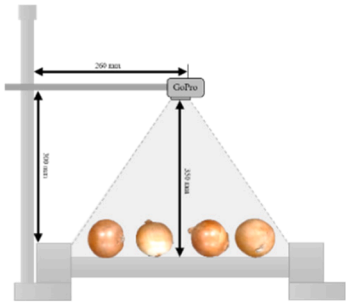


Fig. 8. Different roller situations in specific area A and roller ID infer.

(a) Data collection setup in Saga.

(b) Data collection setup in Yurihonjō.



(c) Camera configuration.

Fig. 9. The onion data collection.

Table 5
The Dataset distribution for training, validation, and testing in Yurihonjō.

	Training set	Validation set	Test set
Onion Detection	4144	1223	617
Roller Detection	1496	412	216
Anomaly detection (Top View)	13,623	-	289
Anomaly detection (Side View)	18,608	-	189
Anomaly detection (Bottom View)	10,310	-	320
total	46,181	1653	1631

For YOLO training, the dataset was split into 70 % for training, 20 % for validation, and 10 % for testing. As for anomaly detection, the model was trained on the normal onion images, with both normal and abnormal images included in the test set to evaluate its performance. The dataset distribution is illustrated in Table 5. This comprehensive data preparation and annotation process ensured a robust foundation for both object detection and anomaly identification tasks.

3.2. Anomaly detection on the Saga dataset and comparison with classification models

To evaluate the effectiveness of our proposed feature-adaptive weight optimization method, we utilized the Saga dataset, which includes 5000 images of normal onions and 350 images of defective onions. From this dataset, 350 normal onion images were combined with 350 defective onion images as a test set. The remaining 4650 normal onion images were used to train our feature-adapted weight optimization method.

As shown in Fig. 10, the experimental result demonstrated that our method achieved an AUROC of 99.87 %, outperforming the original ST model's performance of 99.61 %. This improvement highlights the effectiveness of our approach in enhancing anomaly detection accuracy through feature-adapted weight optimization.

To evaluate its performance against traditional classification models, we constructed two training configurations using the Saga dataset. In both settings, the test set consisted of 100 onion images (50 normal and 50 abnormal), with most of the abnormal samples representing unseen defect types that were not included in the training data. The first configuration used a balanced training set containing 300 normal and 300 abnormal onions. The second configuration employed an imbalanced training set with 4650 normal onions and 300 abnormal onions.

This second setup reflects a common characteristic of real-world industrial data—normal samples are abundant and easy to collect, whereas abnormal samples are relatively scarce and highly diverse. We

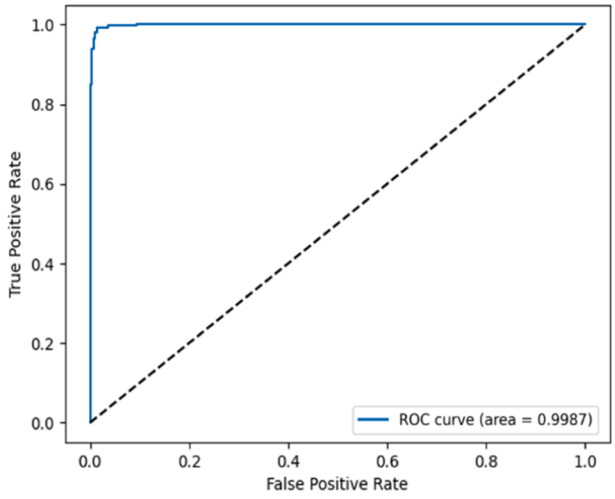


Fig. 10. AUROC curve of the ST model with feature-adaptive weight optimization.

trained several popular CNN-based classification models on both configurations, and the results are summarized in Table 6.

As shown in Table 6, our anomaly detection method achieved an accuracy of 92 % under Configuration 1, which used a balanced training dataset. In Configuration 2, where the training data included a large number of normal samples and fewer abnormal ones, the model achieved the highest accuracy of 99 %, outperforming all CNN-based classification models. In contrast, some classification models, such as VGG16, even experienced a drop in accuracy when trained on the imbalanced dataset. These results demonstrate both the effectiveness of anomaly detection in imbalanced data scenarios and its superior robustness in identifying previously unseen defect types.

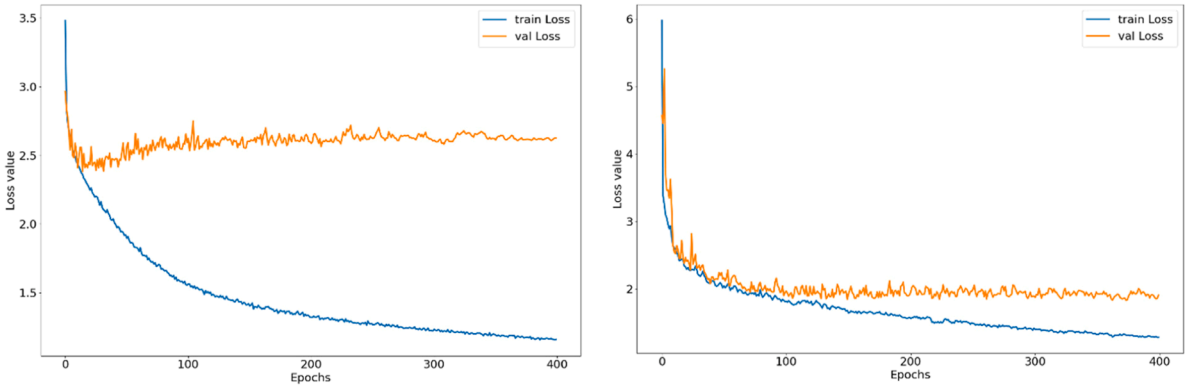
3.3. Results of the model training

3.3.1. The result of YOLOv8s training

The YOLOv8s loss function for detection tasks incorporates three key components: box regression loss, classification loss, and distribution focal loss (DFL). YOLOv8 refines the bounding box regression with the Scaled IoU (SIoU) loss, which accounts for spatial and shape constraints beyond the standard IoU measure to more accurately align predicted boxes with ground-truth objects. The classification loss employs a binary cross-entropy (BCE) strategy, where class probabilities are estimated through sigmoid activation functions and penalized according to their deviation from one-hot encoded ground-truth vectors. The DFL en-

Table 6
Accuracy comparison of classification models on the onion dataset.

	Alexnet	VGG16	Resnet 18	Resnet 101	EfficientNet	Ours
Configuration1	94 %	95 %	95 %	95 %	96 %	92 %
Configuration2	96 %	97 %	93 %	97 %	98 %	99 %



(a) Loss curve of onion detect model. (b) Loss curve of roller detect model.

Fig. 11. The change curve on loss function.

courages more precise localization and effectively captures uncertainty in bounding box predictions. The loss of YOLOv8 is computed as a weighted sum of these three terms, ensuring that the network optimizes both the positioning and classification of detected objects while maintaining robust and fine-grained localization accuracy.

$$Loss_{total} = \lambda_{box} L_{BOX} + \lambda_{cls} L_{CLS} + \lambda_{dfl} L_{DFL} \tag{11}$$

$$L_{BOX} = 1 - \left(IoU - \frac{\rho^2 (b_p, b_{gt})}{c^2} - \alpha v \right) \tag{12}$$

$$L_{CLS} = -\frac{1}{C} \sum_{c=1}^{C} [y_c \log(\hat{p}_c) + (1 - y_c) \log(1 - \hat{p}_c)] \tag{13}$$

$$L_{DFL} = \frac{1}{4} \sum_{d \in (x,y,w,h)} - \left(w_j \log\left(\hat{p}_j\right) + w_{j+1} \log\left(\hat{p}_{j+1}\right) \right) \tag{14}$$

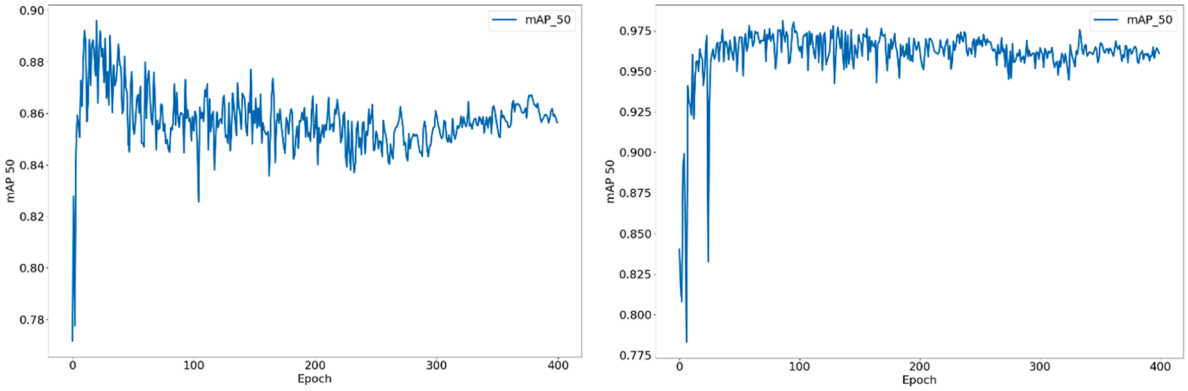
Where IoU is the intersection over union ratio of the predicted box to the real box, b_p represents the predicted box, b_{gt} represents the real box. $\hat{p}_c$ indicates the predicted probability of category, The real class is y_c.

As illustrated in Fig. 11, the training and validation loss curves show consistent decreases across epochs, eventually converging to a stable state. The loss initially drops significantly, reflecting effective learning, and begins to stabilize with minimal fluctuations as the training progresses. However, after a certain point, the validation loss starts to diverge slightly, indicating the onset of overfitting. To prevent this, we selected the epoch corresponding to the highest validation accuracy as the final model checkpoint.

In object detection tasks, mAP50 is used to evaluate model performance, where a higher value indicates more precise object localization. Using the aforementioned training strategy, the YOLOv8s model achieved 87.7 % mAP50 on the onion dataset and 97.5 % mAP50 on the roller dataset. The corresponding mAP50 curves are shown in Fig. 12.

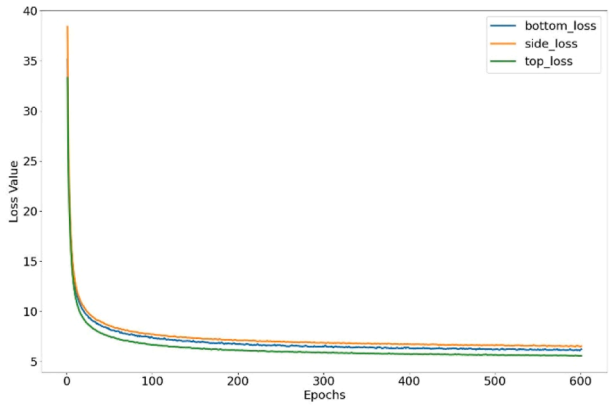
3.3.2. *Training results and threshold optimization for ST models*

As shown in Fig. 1, onion images are divided into three distinct views: top, side, and bottom. Each view is processed using a

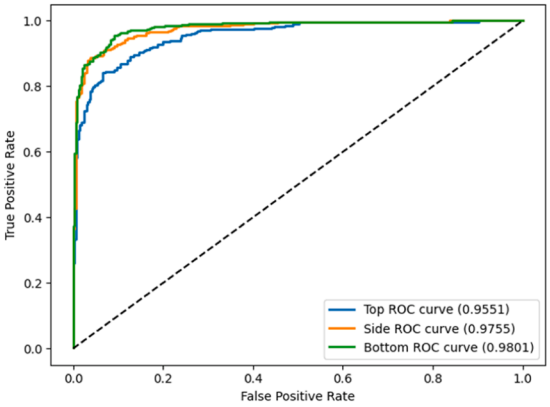


(a) mAP50 of onion detect model. (b) mAP50 of roller detect model.

Fig. 12. The accuracy curve.



(a) Loss curve for three ST models.



(b) AUROC curve for three ST models.

Fig. 13. Loss curve and ROC curve of 3 anomaly detection models.

corresponding ST model ST-top, ST-side, or ST-bottom), specifically designed to extract features and determine onion status. According to Equation (6), the loss function of the feature-adaptive ST model contains two loss terms, the Euclidean distance loss (*L_var*) and the directional similarity loss (*L_dir*). *L_var* minimizes the Euclidean distance between the activation outputs of the student model and the teacher model at each feature block. *L_dir* encourages the feature representation of the student model to always be in the same direction as the teacher model's feature representation.

As shown in Fig. 13(a), all loss values decrease rapidly in the first 100 epochs, which indicates significant learning progress and rapid adaptation by the models. From epoch 100 to 600, the losses tend to stabilize, indicating that the models have converged and entered a steady state of performance improvement. The AUROC curve is shown in Fig. 13(b). The three ST models exhibit AUROC values close to 1, demonstrating strong anomaly detection capabilities. In particular, the Side ST model achieved 0.9551 ROC, the Top ST model achieved 0.9755, and the Bottom ST model achieved 0.9801, highlighting its outstanding classification performance for the given category.

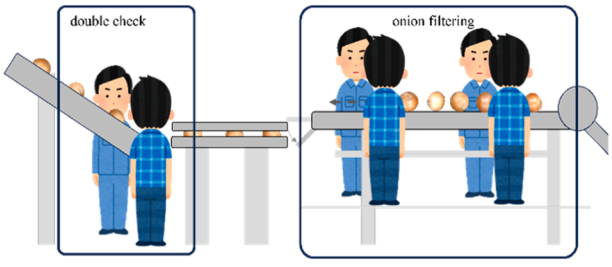
Based on the AUROC curves shown in Fig. 13(b), we determined an optimal threshold for each of the three ST models individually. While higher thresholds typically increase the true positive rate, they also lead to a higher false positive rate. Conversely, setting a lower threshold can reduce FPR but may result in missed anomalies due to a drop in TPR. Therefore, the threshold was selected by balancing these two factors, aiming to maintain a high TPR while minimizing FPR as much as possible. This trade-off allows the model to accurately detect defective onions while keeping the number of false positives at a manageable level.

3.4. Field performance of the onion inspection system

To evaluate the real-world performance of the proposed onion inspection system, we conducted a series of experiments in an industrial onion processing facility. These experiments assessed the system's capabilities across three key aspects: anomaly detection accuracy, filtering accuracy, and overall operational performance in real-world conditions. The evaluation process included comparisons with human operators and measurements of the system's efficiency in identifying and filtering defective onions. Additionally, the system's performance was evaluated in conjunction with the double-check process to determine its effectiveness in enhancing filtering accuracy and reducing reliance on manual labor. The workflow of the onion processing is illustrated in Fig. 14.

3.4.1. Performance of anomaly detection

The proposed system was evaluated for its ability to accurately detect abnormal onions under real-world conditions. Given that defective regions on an onion may be obscured in certain camera frames, the system adopts a conservative approach: if an onion is identified as defective in any frame, it is classified as abnormal; otherwise, it is categorized as normal.



(a) Illustration of the roles of different workers.

(b) Real-world workflow of the onion process.

Fig. 14. Workflow of the onion filtering in the processing in real site.

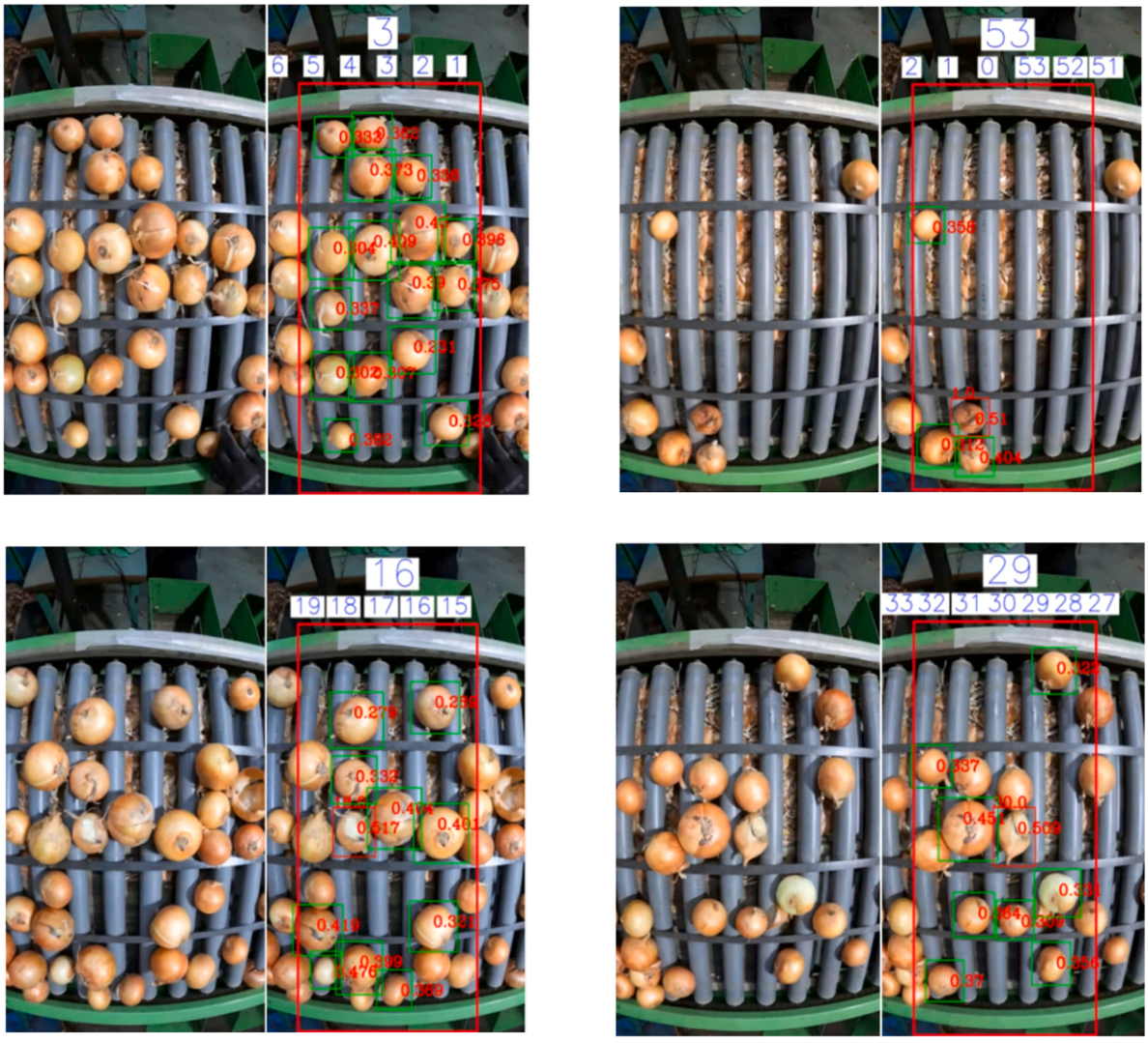


Fig. 15. The detection results on real site.

Table 7
Onion anomaly performance analysis of proposed system on real-site.

	Predicate normal	Predicate abnormal	Total	Acc
Actual normal	1818 (TP)	281(FN)	2099	86.61 %
Actual abnormal	23 (FP)	114 (TN)	137	83.21 %
Total	1841	395	2236	86.37 %

As shown in Fig. 15, the left image illustrates an onion captured by the GoPro camera, and the right image displays the corresponding detection results. In the right image, the large number at the top represents the ID number obtained via serial communication from the DOFM. The ID was assigned to the corresponding conveyor roller using the precise roller ID mapping algorithm, and the IDs for other rollers were inferred. The red bounding boxes in the image indicate the detection regions.

To minimize distortion caused by the GoPro camera lens, the central area with minimal distortion is selected as the primary detection region. While a higher camera position can reduce lens distortion, it compromises the capture of fine surface textures, negatively affecting detection accuracy. Although distortion calibration algorithms could address this issue, they are computationally expensive. Considering the trade-off

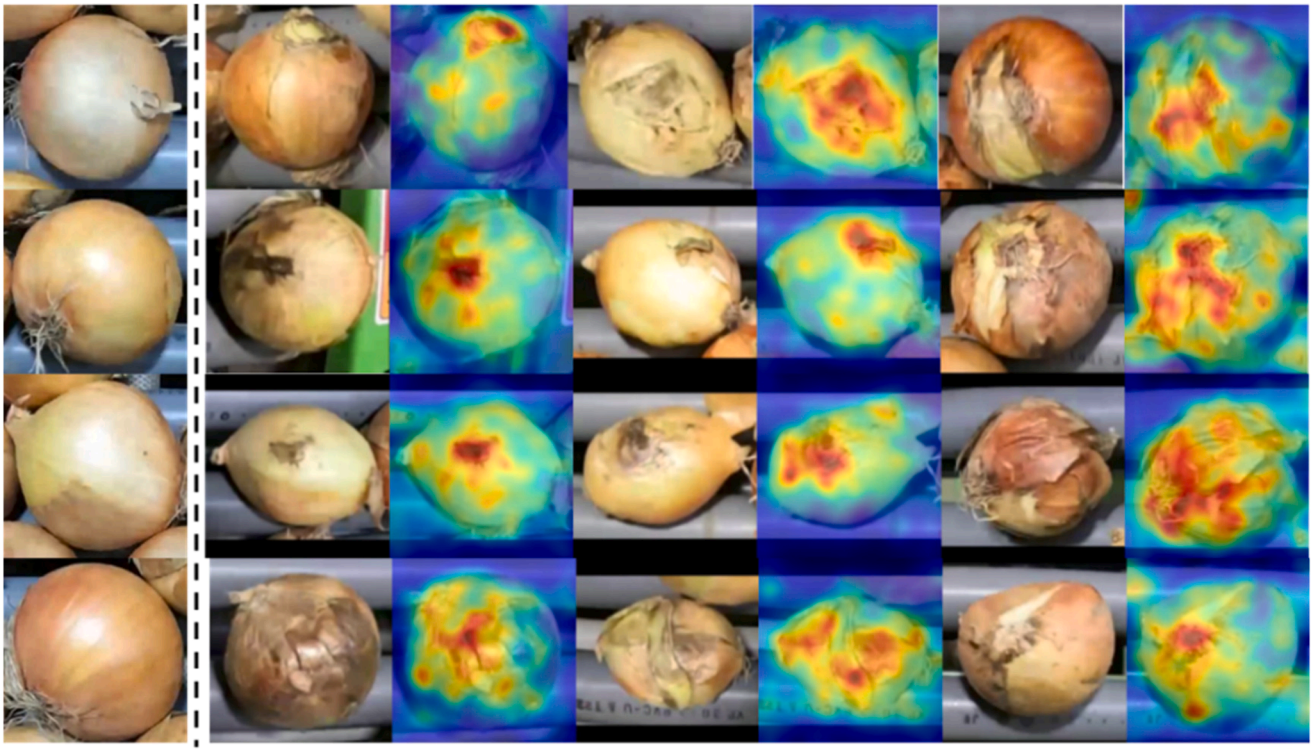


Fig. 16. The visualization of the proposed method on onion.

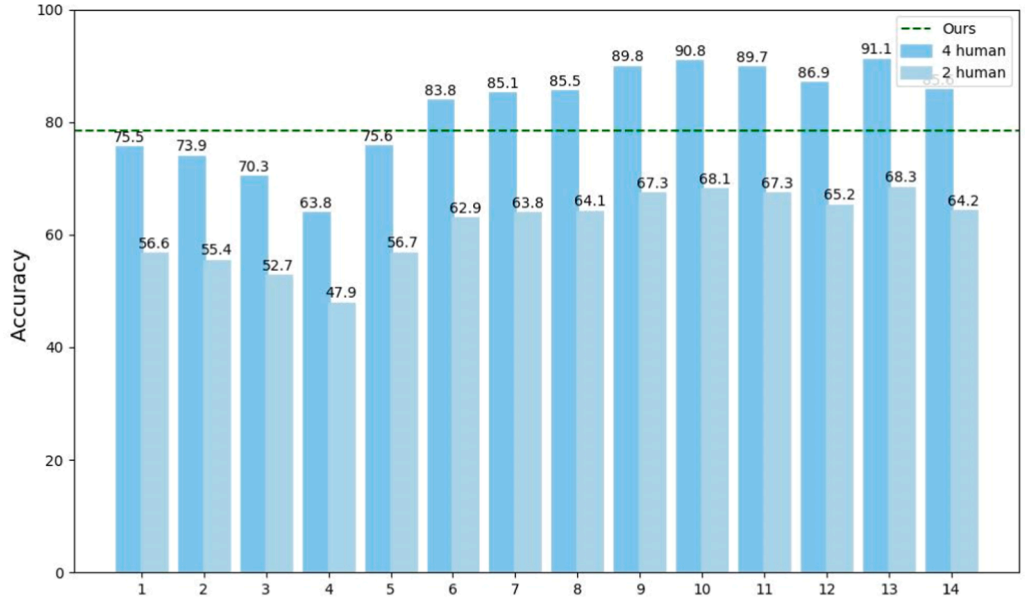
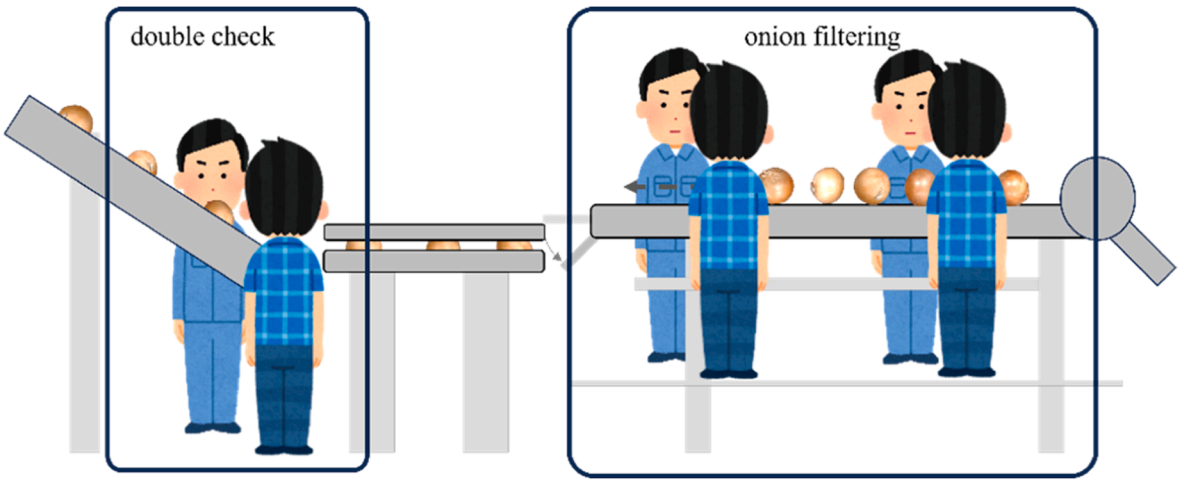
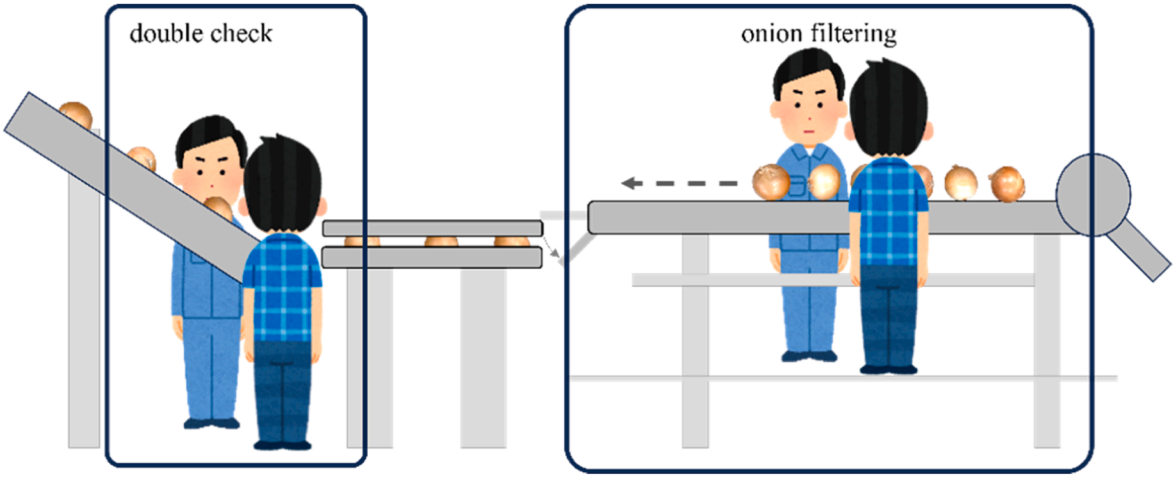


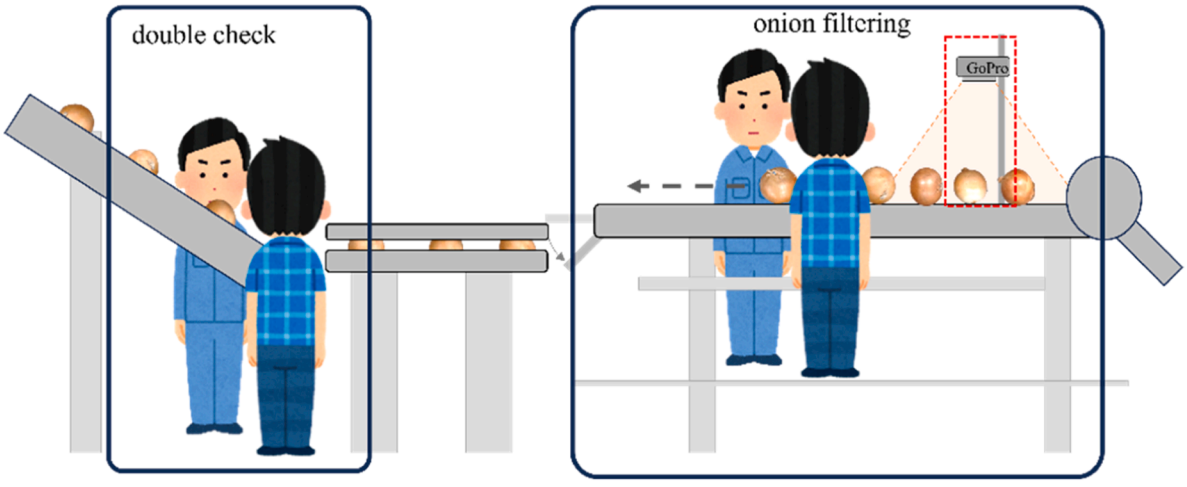
Fig. 17. Filtering performance comparison.



(a) 4 workers scenario.



(b) 2 workers scenario.



(c) 2 workers + AI scenario.

Fig. 18. Illustration of three scenarios.

Table 8
The final check ACC of different onion filtering tests.

	4 workers	2 workers	2 workers + AI
1300 kg onions	24.5 %	43.4 %	23.8 %
1300 kg onions	26.1 %	44.6 %	34.7 %
600 kg onions	24.5 %	43.4 %	26.0 %
300 kg onions	16.2 %	37.2 %	19.0 %
Mean	22.83 %	42.15 %	25.88 %

between image quality and real-time performance, the current detection region was chosen as the optimal solution.

During the detection process, the system marks each detected onion with a rectangular bounding box and displays its anomaly value at the center of the box. Anomalies are classified based on a threshold: onions with anomaly values below the threshold are considered normal and highlighted with green bounding boxes, while those exceeding the threshold are marked defective with red bounding boxes. This annotation method enhances the clarity of detection results, allowing users to intuitively and quickly distinguish between normal and defective onions. This significantly improves the efficiency and practicality of the onion inspection system.

Table 7 summarizes the performance of the onion inspection system. The accuracy is determined based on the system's ability to correctly identify good onions as normal and defective onions as abnormal. The proposed system reaches an accuracy of 86.37 %, The results highlight the system's effectiveness.

Fig. 16 presents representative results from the defect detection process using specialized feature-adaptive ST models designed for anomaly detection at different postures. The left side of the figure displays the original onion images, while the right side presents heatmaps overlaid using class activation maps. In the heatmaps, deeper colors highlight regions with higher importance for the model's interest area by applying a threshold to CAM values, the system identifies and highlights key regions contributing to the detection task. These visualizations demonstrate that, in most cases, the actual defective regions on onions strongly align with the predicted regions, showcasing the robustness and accuracy of the proposed model.

3.4.2. Onion filtering accuracy

To evaluate the filtering accuracy of the proposed system, we analyzed the results of the system's performance during real-world operation. Specifically, the filtering accuracy was calculated as the percentage of abnormal onions correctly identified and filtered out of the total onions processed by the system. Based on this approach, the system achieved a filtering accuracy of 78.5 %.

To benchmark the system's filtering performance, we conducted comparative tests with human operators performing the same task. As illustrated in Fig. 17, the average filtering accuracy of four human operators was 80.81 %, slightly surpassing the system's performance. However, the system surpassed the accuracy of two human operators, showcasing its potential as a reliable alternative for large-scale sorting tasks. Unlike human operators, the system maintains consistent accuracy over extended periods of operation, unaffected by fatigue or human error.

While the system achieved high filtering accuracy, a gap exists between the anomaly detection results and the filtering accuracy observed in real-world operations. One contributing reason is the close spacing of onions on the rollers, which often causes multiple onions to fall together during the filtering process, leading to misclassification of normal onions as abnormal. Additionally, the continuous motion of the roller introduces positional variability, with onions shifting and rotating, as they move through the system. These dynamic changes complicate both the detection and filtering processes, ultimately impacting the system's overall performance.

Table 9
Time consumption for each Task.

Task	YOLO detection	Anomaly detection	Overall processing
Time Range	40ms-50ms	15ms-20ms	55ms-450ms

3.4.3. Validation of double check process in real site

In the onion factory, a double-check process is commonly implemented to ensure the quality of onions before packaging and shipping. This step is particularly important because it acts as the final layer of quality control, identifying and correcting any filtering errors made during the initial inspection process. Maintaining the accuracy of the double check is critical for reducing defective products and upholding product standards. We used the results of the final check to evaluate the performance of the onion filter ability. The final check accuracy is calculated as follows:

$$\text{Final check ACC} = \frac{\textit{bad onions in double check}}{\textit{bad onions in onion filtering} + \textit{bad onions in double check}} \quad (15)$$

Here, a lower final check value indicates higher filtering accuracy, as it reflects lower errors made during the onion filtering process.

As illustrated in Fig. 18, we conducted experiments under three scenarios: four workers performing the filtering manually, two workers performing the filtering manually, and two workers collaborating with the proposed onion inspection system. Table 8 presents the results for various onion filtering.

The results reveal that the setup with two workers and the onion inspection system consistently achieved lower final check values compared to two workers. Specifically, the mean final check accuracy for the "2 workers + AI" setup was 25.88 %, significantly outperforming the 42.15 % achieved by two workers alone and nearly matching the performance of four workers (22.83 %). These results demonstrate that the proposed system not only reduces the reliance on manual labor but also enhances filtering accuracy.

3.4.4. Performance metrics in real-time inspection detection

During system operation, multiple processes run in parallel to ensure efficient detection. As shown in Table 9, the YOLO detection is completed within 40–50 ms and anomaly detection requires 15–20 ms, enabling efficient and seamless operation. On average, the camera's detection area contains 13 onions under normal conditions, but in high-load scenarios, it can accommodate up to 30 onions simultaneously. The number of onions directly impacts processing time, with the overall detection period ranging from 55 ms to 500 ms, corresponding to a processing rate of approximately 2–20 FPS. The system operates at a rate of 4 FPS on average. Despite this variability, our approach shows the ability to handle up to 30 onions in the same frame without compromising accuracy or speed. This scalability highlights the robustness of our system under fluctuating workload conditions.

In the factory environment, each onion typically remains within the camera's field of view for approximately 1.5 s. Even under challenging conditions, such as when a large number of onions are present simultaneously in the detection area, the processing time for a single frame increases due to the higher computational demand. The onions can still be detected from multiple angles, usually more than three times. This repeated detection not only improves accuracy but also ensures reliable identification of defective onions, providing an additional layer of robustness to the system.

3.5. Limitations

Although the proposed onion inspection system with a novel feature-adaptive weight optimize method for anomaly detection can significantly reduce the factory's dependence on manual labor, certain

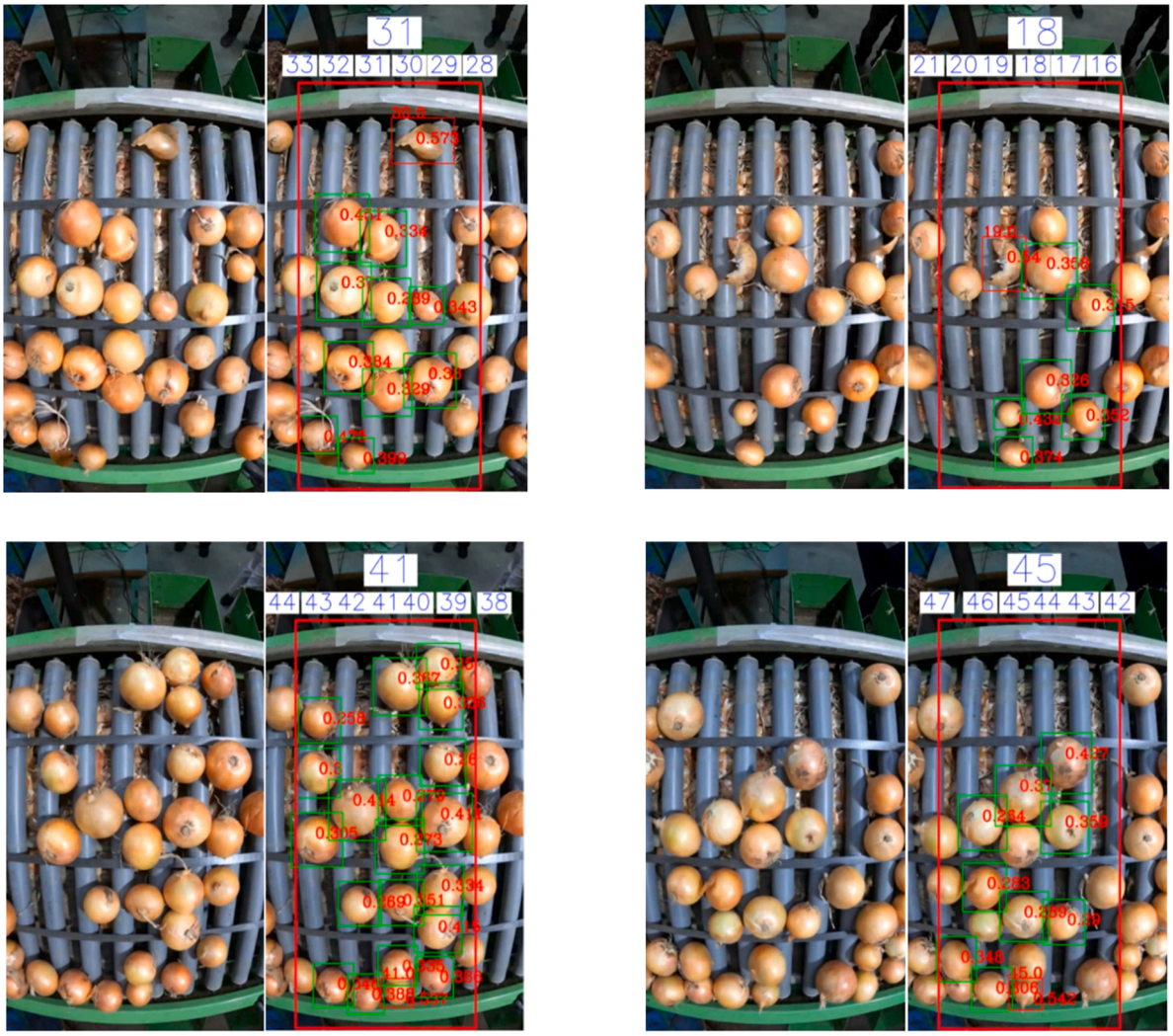


Fig. 19. Failure case on onion detect.

limitations remain.

First, the system's performance heavily depends on the detection capabilities of the YOLO model. The accuracy of the YOLO model directly impacts subsequent detection processes. During operation, onions often lose portions of their outer skins as they are transported and rolled on the conveyor rollers. As shown in Fig. 19, these peeled onion skins can be mistakenly identified as onions. Second, the fixed camera angle introduces challenges related to size variations among onions on the conveyor belt. Larger onions can obstruct smaller ones, resulting in occlusion issues. This can lead to inaccurate image cropping and subsequent misdetections, particularly when multiple onions overlap in the detection area.

Future work includes adopting more robust detection models and optimizing the mechanical design of the inspection system.

4. Conclusions

This study presents an innovative onion inspection system designed to achieve precise anomaly detection and efficient onion filtering in industrial applications. The proposed feature-adaptive weight optimization method for anomaly detection enhances sensitivity to deviations while effectively addressing the inherent variability of normal data. Its effectiveness and efficiency have been validated using the collected onion dataset, demonstrating superior performance compared to baseline models. The system employs a collaborative multi-model framework, combining YOLOv8s for state and pose detection with the anomaly detection model for defect identification, achieving a sorting accuracy. By integrating these detection results, the system performs real-time, autonomous filtering, significantly reducing labor requirements. Moreover, the proposed inspection system can be adapted to other round-shaped fruits such as apples and oranges, highlighting its versatility and potential for broader applications in agricultural automation.

Ethics statement

Not applicable: This manuscript does not include human or animal research.

CRediT authorship contribution statement

Ziwei Song: Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft. **Prawit Buayai:** Data curation, Methodology, Project administration, Software, Supervision, Validation, Writing – review & editing. **Koji Makino:** Investigation, Project administration, Funding acquisition, Validation. **Xiaoyang Mao:** Conceptualization, Funding acquisition, Methodology, Supervision, Resources, Writing – review & editing.

Declaration of competing interest

Xiaoyang Mao reports financial support was provided by Ministry of Agriculture, Forestry, and Fisheries of Japan.

Acknowledgments

This study was supported by a research grant from the Ministry of Agriculture, Forestry, and Fisheries of Japan.

We would like to thank Sojitz YURI-NOJIN Co., Ltd. for supporting our experiments by providing the experimental site and onions, and to the local farmers for their invaluable on-site assistance.

Data availability

The authors do not have permission to share data.

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